ZOO514 -Practical Exam Notes

Generated by Muhammad Kashif(03064963933)

1. What are the two main components that carbohydrates contain?

a) Carbon and oxygen

b) Carbon and hydrogen

c) Oxygen and nitrogen

d) Nitrogen and sulfur

2. Which of the following is NOT a type of carbohydrate?

a) Monosaccharides

b) Dipeptides

c) Polysaccharides

d) Oligosaccharides

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3. Which chemical test is used to differentiate between monosaccharides and polysaccharides? a) Molisch's Test

b) Benedict's Test

c) Iodine Test

d) Biuret Test

4. The Biuret test is used to detect the presence of which class of biomolecules?

a) Lipids

b) Proteins

c) Nucleic acids

d) Carbohydrates

5. Which amino acids contain aromatic groups in their side chains?

a) Glycine and alanine

b) Lysine and arginine

c) Tryptophan and tyrosine

d) Aspartic acid and glutamic acid

6. Fatty acids react with which of the following reagents to produce soap? a) Ethanol

b) Chloroform

c) Acetic acid

d) Sodium hydroxide

7. Which type of epithelial tissue is found in the kidney tubules?

a) Simple columnar

b) Simple cuboidal

c) Simple squamous

d) Transitional

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8. Intercalated discs connect individual cells in which of the following tissue types?

a) Smooth muscle

b) Skeletal muscle

c) Cardiac muscle

d) Nervous tissue

9. Which component of the blood helps in blood clotting?

a) Monocytes

b) Neutrophils

c) Platelets

d) Eosinophils

10. The process by which cell membranes pull away from the cell wall due to water loss is called:

a) Exocytosis

b) Endocytosis

c) Plasmolysis

d) Phagocytosis

11. Pepsinogen is converted into its active form pepsin in which conditions? a) Basic pH

b) Low temperature

c) Acidic pH

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d) Denaturing conditions

12. Which species is found in tropical rainforests, montane forests, and swamp forests in Central and West Africa?

a) Mountain hawk-eagle

b) Kangaroo

c) Polar bear

d) Chimpanzee

13. What is the primary food source of polar bears?

a) Whales

b) Seals

c) Fish

d) Reindeer

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14. Where do female polar bears make their dens while giving birth? a) On land

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b) On sea ice

c) Both on land and sea ice

d) In the ocean

15. Which of the following best describes the social structure of chimpanzees?

a) Solitary

b) Pairs

c) Complex communities with multiple males, females and offspring

d) Randomly mixed groups

16. Slides showing spindle-shaped muscle cells with a single central nucleus represent which tissue? a) Skeletal muscle

b) Cardiac muscle

c) Smooth muscle

d) Elastic cartilage

17. What is the ecological role of adipose tissue in the body?

a) Structural support

b) Mechanical protection

c) Energy storage

d) Transport

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18. Which type of prepared microscope slide would show rows of small chondrocytes forming a striped pattern? a) Hyaline cartilage

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b) Adipose tissue

c) Fibrocartilage

d) Bone tissue

19. Intervertebral discs are made up of which type of specialized connective tissue?

a) Blood

b) Adipose tissue

c) Fibrocartilage

d) Hyaline cartilage

20. In the process of saponification, which components are formed from fats or oils?

a) Hydrocarbons and glycerol

b) Fatty acids and ethanol

c) Glycerol and fatty acid salts

d) Glucose and fructose

21. Which chemical test for carbohydrates involves the formation of a purple ring at the interface of two solutions? a) Benedict's test

b) Biuret test

c) Iodine test

d) Molisch's test

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22. Simple cuboidal epithelium is found lining ducts and secretory parts of: a) Large glands

b) Small glands

c) Endocrine glands

d) Exocrine glands

23. Which connective tissue is characterized by an abundant extracellular matrix rich in collagen fibers? a) Blood

b) Adipose tissue

c) Fibrous connective tissue

d) Cartilage

24. The cells of nervous tissue that provide nutritional and developmental support are called: a) Axons

b) Neurotransmitters

c) Myelin

d) Neuroglia

25. Skeletal muscle tissue is made up of tubular, elongated and _____ cells.

a) Branched

b) Flat

c) Multinucleate

d) Cuboidal

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26. Which ecosystem was home to Pan troglodytes (chimpanzees)?

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a) Forest ecosystem

b) Mountain ecosystem

c) Arctic ecosystem

d) Island ecosystem

27. Which species made ground nests for breeding and shelter?

a) Kangaroos

b) Chimpanzees

c) Mountain Hawk-eagles

d) Polar Bears

28. Arrange the steps involved in plasmolysis of blood cells in the correct sequence:

i. Add distilled water ii. Prepare a blood smear on a slide iii. Prick your finger with a blood lancet iv. Observe cell swelling under a microscope

a) i, ii, iii, iv

b) iii, ii, i, iv

c) iii, iv, ii, i

d) ii, iii, i, iv

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29. Which enzyme breaks down proteins into smaller peptides in the stomach?

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a) Lipase

b) Amylase

c) Nuclease

d) Pepsin

30. What are the elongated claw-like structures called by which chimpanzees swing from branches? a) Pincers

b) Talons

c) Claws

d) Tarsi

31. Which of the following amino acids does NOT react readily with ninhydrin reagent?

a) Lysine

b) Arginine

c) Alanine

d) Phenylalanine

32. During tissue repair, fibroblasts are most likely to differentiate into:

a) Smooth muscle cells

b) Skeletal muscle cells

c) Fat cells

d) Collagen fibers

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33. Which of the following stains could be used to visualize different blood cells in a smear under a microscope? a) Molisch's reagent

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b) Ninhydrin solution

c) Wright's stain

d) Biuret reagent

34. The epidermis of skin is made up of which type of epithelial tissue?

a) Simple squamous epithelium

b) Transitional epithelium

c) Simple cuboidal epithelium

d) Pseudostratified columnar epithelium

35. Which cell junctions link cardiac muscle cells together into functional sheets?

a) Tight junctions

b) Adherens junctions

c) Gap junctions

d) Desmosomes

36. Which of the following is a liquid connective tissue?

a) Tendons

b) Bone

c) Blood

d) Cartilage

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37. According to field observations, nesting sites for Mountain Hawk-eagles are typically found:

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a) On cliffs and rocky edges of open areas

b) In dense tropical foliage

c) In the canopy of tall trees

d) On the ground near sources of prey

38. Polar bears rely on sea ice primarily as a:

a) Denning site

b) Basking spot

c) Hunting ground

d) Migratory pathway

39. Which species frequented coastal areas, savannas, and open grasslands in the Australian continent? a) Chimpanzees

b) Polar Bears

c) Mountain Hawk-Eagles

d) Kangaroos

40. Whooping calls and chest beats are unique courtship behaviors exhibited by:

a) Kangaroos

b) Polar bears

c) Chimpanzees

d) Mountain Hawk-Eagles

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Here are the correct options for all the 40 MCQs:

1. **b) Carbon and hydrogen**
2. **b) Dipeptides**
3. **c) Iodine Test**
4. **b) Proteins**
5. **c) Tryptophan and tyrosine**
6. **d) Sodium hydroxide**
7. **b) Simple cuboidal**

8. c) Cardiac muscle

9. c) Platelets

10. c) Plasmolysis

11. c) Acidic pH

12. d) Chimpanzee

13. b) Seals

14. c) Both on land and sea ice

15. c) Complex communities with multiple males, females and offspring

16. c) Smooth muscle

17. c) Energy storage

18. c) Fibrocartilage

19. c) Fibrocartilage

20. c) Glycerol and fatty acid salts

21. d) Molisch's test

22. b) Small glands

23. c) Fibrous connective tissue

24. d) Neuroglia

25. c) Multinucleate

26. a) Forest ecosystem

27. c) Mountain Hawk-eagles

28. b) iii, ii, I, iv

29. d) Pepsin

30. c) Claws

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31. d) Phenylalanine

32. d) Collagen fibers

33. c) Wright's stain

34. a) Simple squamous epithelium

35. c) Gap junctions

36. c) Blood

37. c) In the canopy of tall trees

38. c) Hunting ground

39. d) Kangaroos

40. c) Chimpanzees

41. How many monosaccharide units are typically present in oligosaccharides?

a) 2-5

b) 10-20

c) 3-10

d) Over 100

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42. What is the typical pH range in which pepsin shows optimal activity? a) pH 7-9

b) pH 1-3

c) pH 11-13

d) pH 5-7

43. Approximately how long does it take for a blood smear to completely dry before staining? a) 5 minutes

b) Half an hour

c) 2 hours

d) 10 seconds

44. How many eggs does a female Mountain Hawk-Eagle typically lay at one time?

a) 5-6

b) 1-2

c) 10-12

d) 3-4

45. What is the average weight range of an adult male Chimpanzee in kilograms?

a) 80-100

b) 30-60

c) 150-200

d) 10-25

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46. The process of protein digestion typically begins in which part of the gastrointestinal tract? a) Mouth

b) Esophagus

c) Stomach

d) Small intestine

47. Which type of microscope is required to examine tissue slides and blood smears?

a) Electron microscope

b) Light microscope

c) Fluorescence microscope

d) Confocal microscope

48. Which enzyme plays a key role in the breakdown of lipids? a)

Protease

b) Lipase

c) Lactase

d) Nuclease

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49. The saponification reaction occurs rapidly in which range of temperature?

a) 0-10°C

b) 20-40°C

c) 100-150°C

d) -10-0°C

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50. In smooth muscle tissue, individual cells are approximately how many micrometers wide?

a) 2-5 μm

b) 10-15 μm

c) 50-100 μm

d) 0.2-0.5 μm

51. Which of the following skin structures helps prevent water loss?

a) Sebaceous glands

b) Stratum spinosum

c) Dermis

d) Stratum corneum

52. Which of the following is an incorrect match between tissue type and location?

a) Elastic cartilage – Outer ear

b) Hyaline cartilage – Ends of long bones

c) Fibrocartilage – Lining of stomach

d) Blood – Present within blood vessels

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53. The process by which cells regain turgidity after plasmolysis is called: a) Exocytosis

b) Active transport

c)Bulk transport

d) Deplasmolysis

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54. Which enzyme is secreted as an inactive zymogen prior to activation? a) Nuclease

b) Amylase

c) Pepsinogen

d) Lipase

55. Nests constructed by Mountain Hawk-Eagles are beneficial primarily for which purpose? a) Food storage

b) Hunting smaller prey

c) Safe sites for eggs and chicks

d) Protection from predators

56. Which visible bands seen in light microscopy help distinguish skeletal muscle from smooth muscle?

a) Z-lines

b) Intercalated discs

c) Striations

d) Sarcomeres

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57. Which type of tissue has both a solid and liquid matrix?

a) Nervous tissue

b) Epithelium

c) Bone

d) Dense regular connective tissue

58. Which species is adapted to tree-dwelling with the use of strong tarsal claws?

a) Polar Bears

b) Kangaroos

c) Mountain Hawk-eagles

d) Chimpanzees

59. The Mountain Hawk-Eagle belongs to which order of birds?

a) Struthioniformes

b) Galliformes

c) Passeriformes

d) Accipitriformes

60. Which species of bears has white-colored fur adapted for camouflage in snow?

a) Grizzly Bears

b) American Black Bears

c) Sloth Bears

d) Polar Bears

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Here are the correct options for the 10 numerical MCQs (41-50) and 10 other MCQs (51-60):

41. c) 3-10 42. b) pH 1-3 43. a) 5 minutes

44. b) 1-2

45. a) 80-100 kg

46. c) Stomach

47. b) Light microscope

48. b) Lipase

49. b) 20-40°C

50. a) 2-5 µm

51. d) Stratum corneum

52. c) Fibrocartilage – Lining of stomach

53. d) Deplasmolysis

54. c) Pepsinogen

55. c) Safe sites for eggs and chicks

56. c) Striations

57. c) Bone

58. d) Chimpanzees

59. d) Accipitriformes

60. d) Polar Bears

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Subjective Part

1. **What are the two processes that occur when a plasmolyzed cell is placed in a hypotonic solution?**

Answer: The two processes that occur are:

1) Absorption of water back into the cell

2) Cell regaining its turgid state

2. **Name the slide preparation technique used to observe blood cells under a microscope.**

Answer: Blood smear technique is used to prepare slides for observing blood cells under a microscope.

3. **Which type of cartilage provides the strongest form of connective tissue in the body?**

Answer: Fibrocartilage provides the strongest form of connective tissue in the body.

4. **Which organ produces the inactive precursor of the protein-digesting enzyme pepsin?**

Answer: Pepsinogen, the inactive precursor of pepsin, is produced by the chief cells in the gastric glands of the stomach.

5. **What are the main threats to chimpanzee habitats in the wild?**

Answer: The main threats to chimpanzee habitats are deforestation, logging, mining, and agricultural expansion.

6. **What aspect of the Mountain Hawk-Eagle's vision helps it effectively hunt prey?**

Answer: The Mountain Hawk-Eagle has excellent vision which aids in spotting prey from great distances, helping it effectively hunt.

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7. **Why are female polar bears dependent on snowdrift dens?**

Answer: Female polar bears are dependent on snowdrift dens for giving birth and rearing their young.

8. **How does the hopping posture of kangaroos help them thrive in the wild?**

Answer: The hopping posture allows kangaroos to cover long distances efficiently in search of food, water and suitable breeding sites.

9. **What solution is used to induce plasmolysis in blood cells under a microscope?**

Answer: A 3% NaCl (hypertonic) solution is used to induce plasmolysis in blood cells under a microscope.

10. Which type of muscle tissue shows branching striations under a microscope?

Answer: Cardiac muscle tissue shows complex and branching striations under a microscope.

11. Which organ produces the precursor to pepsin called pepsinogen?

Answer: Pepsinogen is produced by the chief cells in the stomach's gastric glands.

12. Name two types of forest habitats inhabited by chimpanzees.

Answer: Chimpanzees inhabit tropical rainforests and montane forests.

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13. What aspect of nest site selection is crucial for Mountain Hawk-Eagles?

Answer: The location and placement of nests high up in tall canopy trees is a crucial aspect of nest site selection by Mountain Hawk-Eagles.

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14. Why does the loss of sea ice pose a major threat to polar bears?

Answer: Sea ice is critical polar bear habitat for hunting, breeding and denning. Loss of sea ice poses risks to their access to food and ability to successfully rear young.

15. How does the specialized digestive system of kangaroos help them thrive in diverse habitats?

Answer: Kangaroos have a specialized digestive system to extract nutrients from tough and fibrous plant material, helping them thrive across habitats.

16. What are the components of blood that can be observed under a microscope from a prepared slide?

Answer: The components of blood that can be observed under a microscope from a prepared slide are: erythrocytes, platelets, monocytes, lymphocytes and neutrophils.

17. Which epithelial tissue is involved in secretion and absorption processes in the body?

Answer: Simple cuboidal epithelial tissue is involved in secretion and absorption processes in the body.

18. Name the digestive enzyme that breaks down proteins into smaller peptides.

Answer: Pepsin is the digestive enzyme that breaks down proteins into smaller peptides.

19. What ecological factor influences the distribution and behavior of chimpanzees in their forest habitat?

Answer: The availability of food sources like fruits, leaves, seeds and flowers significantly influences the distribution and behavior of chimpanzees in their forest habitat.

20. Why do Mountain Hawk-Eagles typically construct large stick nests high up in trees?

Answer: They construct large stick nests high up in tall trees to provide a safe platform for their eggs and chicks.

21. What is the most significant threat to the survival of polar bears?

Answer: Climate change leading to shrinking sea ice is the most significant threat to the survival of polar bears.

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22. Which solutions are used to demonstrate plasmolysis and deplasmolysis in blood cells?

Answer: A 3% NaCl (hypertonic) solution is used to demonstrate plasmolysis and a distilled water (hypotonic) solution is used to demonstrate deplasmolysis in blood cells.

23. Which connective tissue comprises the middle layer of the heart?

Answer: Cardiac muscle tissue comprises the thick middle layer of the heart.

24. What is the role of hydrochloric acid in the activation of pepsin?

Answer: Hydrochloric acid creates an acidic environment in the stomach which converts inactive pepsinogen into active pepsin.

25. Name two important resources chimpanzees obtain from the plants in their habitat.

Answer: Chimpanzees obtain food resources like fruits, leaves, seeds, and flowers from plants in their habitats.

26. What physical features of Mountain Hawk-Eagle habitat aid their hunting technique?

Answer: Cliffs, rocky edges and open areas aid the hunting technique of Mountain Hawk-Eagles.

27. Where do female polar bears dig dens to birth and raise cubs?

Answer: Female polar bears dig dens in snowdrifts on sea ice or land to birth and raise cubs.

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28. What solution is used to demonstrate deplasmolysis in plasmolyzed blood cells?

Answer: Distilled water (hypotonic solution) is used to demonstrate deplasmolysis in plasmolyzed blood cells.

29. Which muscle tissue type shows spindle-shaped cells with centrally located nuclei?

Answer: Smooth muscle tissue shows spindle-shaped cells with centrally located nuclei.

30. Which enzyme breaks down proteins in an acidic environment?

Answer: Pepsin is the enzyme that digests proteins in an acidic environment.

31. What causes the Shrinkage of plant and animal cells when placed in a hypertonic solution?

Answer: The shrinkage of plant and animal cells when placed in a hypertonic solution is caused by the process of plasmolysis, which is the loss of water from the cell.

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32. Which tissue type acts as a protective barrier and helps in absorption in the digestive and respiratory tracts?

Answer: Columnar epithelial tissue acts as a protective barrier and helps in absorption in the digestive and respiratory tracts.

33. Which digestive enzyme breaks down peptides into individual amino acids?

Answer: Peptidases are the enzymes that break down peptides into individual amino acids.

34. What behaviors exhibited by chimpanzees depend on their complex social structure?

Answer: Complex social behaviors like hierarchical social interactions, territorial displays, cooperative hunting and child rearing depend on the complex social structure of chimpanzee communities.

35. What physical adaptations help the Mountain Hawk-Eagle effectively hunt prey?

Answer: Sharp talons to catch prey combined with excellent long-distance vision aid the Mountain Hawk-Eagle in effectively hunting prey.

36. Why are monitoring programs important for the conservation of polar bears?

Answer: Monitoring programs help scientists track changes in polar bear distribution, habitat use and population numbers over time, which guides conservation efforts.

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37. What causes the cells to regain their normal shape in the process of deplasmolysis?

Answer: The absorption of water back into the cells causes them to regain their normal turgid shape in the process of deplasmolysis.

38. Which tissue type comprises the outermost protective layer of the skin?

Answer: Simple squamous epithelial tissue comprises the outermost protective layer of skin.

39. Which enzyme is activated by hydrochloric acid in the stomach during digestion?

Answer: Pepsinogen is activated by hydrochloric acid to form the active enzyme pepsin which participates in protein digestion in the stomach.

40. What ecological roles do chimpanzees play in their forest habitats?

Answer: Chimpanzees play roles as seed dispersers, pollinators and prey species, influencing forest regeneration and food web dynamics.

41. What courtship displays performed before mating season strengthen the bond between Mountain Hawk-Eagles?

Answer: Elaborate aerial displays, calls and food exchanges performed before mating season help strengthen pair bonds between Mountain Hawk-Eagles.

42. What efforts can help mitigate the impacts of climate change on polar bears?

Answer: Efforts like reducing global greenhouse gas emissions, preserving critical habitat and monitoring population changes can help mitigate climate impacts on polar bears.

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43. What causes swelling and bursting of RBCs when placed in hypotonic solution?

Answer: The unbalanced diffusion of water molecules into the cell causes swelling and bursting of RBCs when placed in a hypotonic solution.

44. Which muscle tissue type shows striations under a microscope?

Answer: Skeletal and cardiac muscle tissues show visible alternate light & dark band striations under a microscope.

45. Which enzyme acts in the acidic pH of the stomach to digest proteins?

Answer: Pepsin is the enzyme activated under acidic stomach pH that participates in the digestion of dietary proteins.

Here are 5 comparative and 5 quantitative questions generated from the file:

1. **Compare the structure of simple squamous epithelium with that of simple columnar epithelium.**

Answer: Simple squamous epithelium consists of a single layer of flat, irregularly shaped cells. Simple columnar epithelium consists of a single layer of tall, tightly packed cells with nuclei close to the basement membrane.

2. **Compare the striations observed in skeletal muscle tissue with that seen in cardiac muscle tissue.**

Answer: Skeletal muscle tissue shows regular, parallel striations, while cardiac muscle tissue shows more complex, branching striations.

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3. **Compare the habitat of chimpanzees with that of Mountain Hawk-Eagles.**

Answer: Chimpanzees inhabit tropical rainforests and montane forests which have a dense canopy and undergrowth. Mountain Hawk-Eagles inhabit mountainous regions including forests and areas with cliffs, rocky edges and open spaces.

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4. **Compare the threats faced by chimpanzee populations with those faced by polar bears**.

Answer: Chimpanzees are threatened mainly by habitat loss due to deforestation, mining etc., while polar bears are most threatened by climate change and melting sea ice in the Arctic.

5. **Compare the process of plasmolysis with that of deplasmolysis.**

Answer: Plasmolysis involves cell shrinkage due to water loss in a hypertonic solution. Deplasmolysis involves regained cell shape and size as water re-enters the cells in a hypotonic solution.

Quantitative Questions:

1. **What is the optimum temperature for the enzymatic activity of pepsin during protein digestion?**

Answer: Pepsin shows optimal enzymatic activity at around 37°C.

2. **What is the approximate size range of blood platelets observed under a microscope?**

Answer: The size range of platelets observed under a microscope is approximately 1.5 – 3 μm.

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3. **What is the typical group size range of chimpanzee communities in their natural habitat?**

Answer: Chimpanzee communities can range from 15 to over 100 members.

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4. **What is the approximate wingspan range of the Mountain Hawk-Eagle?**

Answer: The Mountain Hawk-Eagle has a wingspan ranging from approximately 4 to 6 feet.

5. **What pH levels are required for optimal activation of pepsin enzyme from pepsinogen?**

Answer: Pepsin demonstrates optimal activation from pepsinogen at pH levels between 1.5 to 2.

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